

Use of molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor, in chronic kidney disease-associated anemia in cats

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Abstract

Abstract

Background

Erythropoietic effects of molidustat, a hypoxia-inducible factor prolyl hydroxylase (HIF-PH) inhibitor, were previously demonstrated in healthy cats.

Objective

To evaluate the safety and erythropoietic effects of daily PO administration of molidustat in anemic cats with chronic kidney disease (CKD).

Animals

Twenty-one client-owned CKD cats (4-17 years old) with anemia.

Methods

Multicenter field study; randomized, masked, and placebo-controlled. Cats were treated PO once daily for 28 days with suspensions of control product (CP; n = 6) or 5 mg/kg of molidustat (n = 15). Hematocrit (HCT) was evaluated at weekly intervals. Individual cat treatment success was defined as a $\geq 4\%$ point increase in HCT compared to baseline.

Results

Control group mean HCT remained low throughout the study (20.1%-23.4%). Mean HCT of molidustat-treated cats increased weekly, and a significant increase compared to baseline (23.6%) was first observed on Day 21 (27.3%; $P < .001$; 95% confidence interval [CI], 1.69-5.67). Compared to CP group, mean HCT was significantly higher on Day 21 (27.3% vs 20.1%; $P < .001$; 95% CI, 2.91-10.75) but not significantly higher on Day 28 (27.8% vs 23.4%; $P = .06$; 95% CI, -0.23 to 9.88). The number of individual treatment successes on Day 28 was higher among remaining molidustat-treated cats (7/14) compared to remaining control cats (1/5), but there was no significant difference between groups.

Conclusions and Clinical Importance

Daily PO molidustat administration may stimulate a clinically relevant erythropoietic response in anemic cats with CKD. This HIF-PH inhibitor may be an alternative for managing anemia in cats compared to recombinant EPO treatment.

Abbreviations

adverse event

autoregressive

confidence interval

chronic renal disease

control product

erythropoietin

erythropoietin stimulating agent

Europe

hematocrit

mercury

hypoxia-inducible factor

hypoxia-inducible factor prolyl hydroxylase

renal erythropoietin-producing

repeated measures analysis of covariates

visit

1

INTRODUCTION

Anemia associated with chronic kidney disease (CKD) develops in 30%-65% of CKD cats.¹ Current erythropoietin (EPO) replacement treatment can be associated with life-threatening complications in some cats.^{1 2 3}

Production of EPO is controlled by hypoxia-inducible factor (HIF), found in specialized renal EPO-producing (REP) cells, and degraded under normal oxygen conditions by HIF prolyl hydroxylase (HIF-PH) enzymes, giving mammals the ability to adapt to environmental changes in oxygen levels.^{4 5 6 7} In CKD cats, decreased metabolic activity of failing kidneys leads to relative hyperoxia, decreasing EPO production by the degradation of HIF by HIF-PH enzymes, despite a preserved EPO control mechanism.^{4 7 8} It was therefore postulated that administration of a HIF-PH inhibitor, such as molidustat sodium, may stimulate production of erythrocytes in cats. The erythropoietic effect of daily PO administrations of 5 mg/kg molidustat on hematological parameters was first investigated in healthy adult cats showing an increase in EPO concentration and subsequent increase in hematocrit (HCT) despite normoxic conditions (internal communication). Molidustat administration was well tolerated in study cats.

Considering these results, our aim was to evaluate the ability of molidustat to stimulate production of erythrocytes in client-owned anemic cats with CKD using quantitative assessments of HCT responses over time.

2

MATERIALS AND METHODS

2.1

Study design

Ours was a multicenter, randomized, double-masked, parallel, placebo-controlled field study. It was conducted in accordance with guidelines of the Animal Care Review Committee of Bayer Animal Health (Protocol #205375) and with informed owner consent. It was designed to investigate the effects and safety of molidustat as a PO suspension in client-owned cats in the United States and Europe with anemia associated with CKD when administered daily for 28 days.

2.2

Inclusion and exclusion criteria

To be eligible for inclusion, cats had to be ≥ 1 year of age with body weight ≥ 2.0 kg, diagnosed with anemia (PCV $\leq 27\%$ at screening and on Day 0) associated with CKD (based on historical patient data and baseline serum creatinine concentration > 1.8 mg/dL without staging), negative urine culture and a systemic blood pressure (SBP) ≤ 165 mm Hg pretreatment. Cats could be receiving crystalloid fluids SC and comorbidities had to be stable for a minimum of 2 weeks or for at least 6 months in the presence of hyperthyroidism.

Cats were excluded if they were intended for breeding, pregnant or lactating, had an untreated urinary tract infection, or were difficult to handle. Cats also were excluded if they had received previous treatment with recombinant EPO or a blood transfusion or both, severe clinical signs of inflammation not attributed to CKD, other uncontrolled clinically relevant medical conditions, had been diagnosed with feline leukemia or feline immunodeficiency virus, or had clinical signs of gastrointestinal bleeding.

2.3

Randomization and masking

All study personnel involved in clinical evaluations, observations, or collection of samples were masked to treatment and study group. Dispensers and any study personnel witnessing or administering treatments were not involved in any other roles. To ensure masking, 2 alphabetical treatment codes were randomly assigned to the molidustat or control product (CP), and cats were randomly assigned to 1 of 2 treatment groups at a ratio of 2:1 (molidustat: CP) based on order of entry into the study. A global randomization form was created using SAS Statistical Software (version 9.3, Cary, North Carolina) and exported to the MEDRIO system (MEDRIO, San Francisco, California). The assignment of cats to study groups was performed in MEDRIO by the dispenser at the study site.

2.4

Treatment phase and primary hematological parameters

Primary clinical endpoints were HCT and PCV responses over time. Each cat qualifying for enrollment underwent a screening evaluation on Day -7 (± 2 days) which included physical examination, SBP and body weight measurements, urinalysis and bacterial culture, routine serum biochemistry and hematology, and PCVs. Starting on Day 0, cats were treated PO once daily by the owner before feeding for at least 28 consecutive days with a suspension of either CP

(vehicle only) or molidustat (25 mg/mL). Molidustat was administered at a dosage of 5 mg/kg, equivalent to 0.2 mL/kg. The same volume of vehicle was administered to CP cats.

Blood was collected to assess hematological parameters with body weights recorded during each visit (V1-V5), which occurred on Days 0 (V1; pretreatment), 7 (V2), 14 (V3), 21 (V4), and 28 (V5) $\pm$ 2. Blood also was collected to evaluate serum biochemistry on Days 0 and 28. Three to 5 SBP measurements (within 20% variation) were recorded and then averaged before blood sampling on Days 0 (pretreatment), 14, and 28 after the cat had a chance to acclimate to a quiet room. Each study site used either an ultrasonic doppler or oscillometric SBP device. A central laboratory was used for evaluation of clinical pathology specimens in the United States (Antech GLP, Morrisville, North Carolina) and Europe (IDEXX GmbH, Ludwigsburg, Germany).

Treatment success was evaluated up to and including Day 28. Any clinical findings were recorded as adverse events (AEs) or serious AEs (causing death or life-threatening condition) and immediately reported to the Investigator whether associated with treatment or not.

2.5

Hematological effect evaluation

Individual treatment success for each cat was defined as the primary endpoints reaching a clinically relevant increase. A clinically relevant increase in hematological parameters was defined as a >25% increase or an absolute 4% point increase or both in PCV or HCT from baseline up to and including Day 28.

2.6

Continuation phase

After the 28-day treatment phase, owners could voluntarily include their cats in a nonplacebo-controlled continuation phase for up to 8 additional weeks (File ^{S1}).

2.7

Statistical analysis

Hematocrit and PCV, measured multiple times during the study including baseline, were analyzed using a repeated measures analysis of covariance (RMANCOVA), with treatment, time (week), and treatment by time terms in the model as fixed effects, and animal identified as subject in the repeated statement (MIXED procedure in SAS). Pretreatment (baseline) values were used as a covariate, and remained in the model, regardless of significance. Random effects were included in the model consisting of the clinic (SITE) and region (Europe or United States). The best covariance structure in the RMANCOVA was investigated and determined to be the autoregressive (1) AR (1). If treatment by time interaction was significant at the .05 level, then treatment group comparisons for each day were performed. If the 2-way interaction was not significant, the main effect of treatment group was evaluated for any significant treatment group effect. All terms in the model were evaluated at an unadjusted alpha = .05. Mean HCT % ($\pm$ SD) over a 28-day treatment phase for cats treated with either CP or molidustat was graphed.

Various treatment success definitions (binary in structure, reported in percentages) were analyzed using 2 methods: a nonparametric Fisher's exact test and a generalized linear mixed model (PROC GLIMMIX, SAS). A binomial distribution was assumed, and a logit link was used, adjusting the degrees of freedom using Kenward Rogers

adjustment and including baseline values as a covariate. The number of treatment successes in each group at each visit from Day 7 up to and including Day 28 were counted and compared to baseline.

Separate analyses of both HCT and PCV values also were performed using a model testing for differences in study day for each treatment group (ie, baseline control analyses). An analysis of covariance with fixed term of Day (0, 7, 14, 21, and 28) and pretreatment baseline (Day 0) as a covariate was used to test for Day differences. All pairwise comparisons of mean HCT and PCV values compared to baseline for each visit were tested at the .05 level of significance.

3

RESULTS

3.1

Study population

From 65 cats screened at 11 American and 9 European sites, 23 client-owned CKD cats with anemia were enrolled in the 28-day study (Figure ¹).

FIGURE 1

Enrollment flowchart of the global study population of the randomized pilot field study evaluating the erythropoietic effects of daily oral administration of molidustat in client-owned anemic cats with chronic kidney disease (CKD). CP, control product; EU, Europe; US, United States.

Baseline characteristics of the study population showed that groups of mostly domestic short-haired cats (n = 20/23) were evenly matched in age and body weight, ranging from 4 to 17 years and 2 to 6 kg, respectively, and nearly evenly distributed by sex (Table ¹).

TABLE 1

Baseline mean characteristics (±SD) of the efficacy study population on Day 0, Visit 1 (n = 21).

Variables	CP (n = 6)	Molidustat (n = 15)
Demographics		
Age (years)	10.6 (5.4)	12.2 (3.8)
Sex (M/F) ^a	1M/5F	8M/7F
Breed (DSH)	6	13
Region (% from US)	33	60
Physical examination		
Body weight (kg)	3.9 (1.26)	3.5 (1.07)
SBP (mm Hg)	147.8 (17.97)	148.0 (24.43)
Laboratory results		

Variables	CP (n = 6)	Molidustat (n = 15)
Blood urea nitrogen (mg/dL) reference interval: 34.2-81	170.4 (57.12)	177.6 (71.52)
Creatinine (mg/dL) reference interval: <1.8	5.0 (2.8)	5.0 (2.8)
HCT%	23.2 (2.50)	23.6 (3.23)
PCV%	21.3 (2.98)	22.5 (2.91)
Hb (g/dL)	6.9 (1.03)	7.49 (1.06) ^b

Abbreviations: CP, control product; DSH, domestic short haired; F, female; Hb, hemoglobin; HCT, hematocrit; M, male; NA, not applicable; SBP, systemic blood pressure; US, United States.

a

All cats were neutered or spayed except for 1 intact female in the CP group.

b

n = 14.

There were no comorbidities unrelated to CKD in the CP group, whereas there were 3 cats with hyperthyroidism, 1 with coccidiosis, 1 with cystitis, and 1 with suspected osteoarthritis in the molidustat-treated group (Table ^A, Supporting Information). Several concomitant medications were used during the study, administered to treat clinical signs of CKD or comorbidities at the discretion of the attending veterinarian (Table ^A, Supporting Information).

3.2

Hematological parameters

Mean ($\pm$ SD) baseline HCT of the CP group (n = 6) was 23.2% ($\pm$ 2.50) and weekly responses remained low (20.1%-23.4%) at all assessed post-treatment timepoints (Figure ²).

FIGURE 2

Mean hematocrit (HCT % $\pm$ SD) values over a 28-day treatment phase with either molidustat (n = 15 to 14 due to missing data on Day 28) or CP (n = 6 to 3 due to missing data on Day 28) of anemic cats with CKD. *Significant difference at $P < .05$ between groups on Day 21 is evident; the 95% CI for the difference is 2.91-10.75. CI, confidence interval; CKD, chronic kidney disease; CP, control product.

Mean HCT of the molidustat-treated group increased weekly and a significant increase compared to baseline (23.6%) was first observed on Day 21 (27.3%; $P < .001$; 95% confidence interval [CI] for the difference, 1.69-5.67; Table ²). Mean PCV significantly increased as early as Day 14 in the molidustat-treated group compared to baseline (24.7% vs 22.5%; $P = .02$). In contrast, no significant increases relative to baseline were observed in the CP group at any timepoint.

TABLE 2

Pairwise comparisons with mean baseline HCT and PCV values ($\pm$ SD) in the molidustat-treated group (n = 15).

Hematological parameter	Visit 1 (Day 0)	Visit 2 (Day 7)	Visit 3 (Day 14)	Visit 4 (Day 21)	Visit 5 ^a (Day 28)
HCT (%)	23.6 ± 3.23	25.2 ± 4.34	25.5 ± 5.94	27.3 ± 4.82	27.8 ± 5.25
Difference from baseline	1.6	1.9	3.7	4.2	
<i>P</i> value	.1	.06	<.001	<.001	
PCV (%)	22.5 ± 2.91	24.2 ± 4.16	24.7 ± 4.95	26.4 ± 4.51	26.2 ± 5.24
Difference from baseline	1.7	2.2	3.9	3.7	
<i>P</i> value	.07	.02	<.001	<.001	

Note: Bold indicates statistically significant values.

Abbreviation: HCT, hematocrit.

^a

n = 14.

Compared to the CP group, mean HCT of the molidustat-treated group was significantly higher on Day 21 (27.3% vs 20.1%; *P* < .001; 95% CI, 2.91-10.75) but was not significant on Day 28 (27.8% vs 23.4%; *P* = .06; 95% CI, -0.23 to 9.88; Figure ²).

The number of individual treatment successes on Day 28 was higher among molidustat-treated cats (7/14) compared to CP cats (1/5; Table ³). However, no significant difference was found between groups. Other hematological parameters such as red blood cell and reticulocyte counts and Hb also increased over time, but statistical analysis was not performed (Table ^B, Supporting Information).

TABLE 3

Percent treatment success defined as a >25% increase and/or a 4% point increase in HCT from baseline (Day 0) up to and including Day 28 in anemic cats with CKD given either molidustat or the control product (CP).

Group	Day 7	Day 14	Day 21	Day 28
CP	16.7% (1/6)	20% (1/5 ^a)	0% (0/5)	20% (1/5)
Molidustat	20% (3/15)	30% (5/15)	46.7% (7/15)	50% (7/14 ^b)
<i>P</i> value				.37

Note: Bold indicates statistically significant values.

Abbreviations: CKD, chronic kidney disease; HCT, hematocrit.

^a

One cat died shortly after Day 7.

^b

One cat was excluded because of potential impact on HCT from dehydration.

3.3

In-use safety

Any clinical findings evident on physical examinations before and after the 28-day treatment period were reported as AEs (Table ⁴). It was possible for a cat to have multiple reported AEs. Biochemistry variables were within reference intervals (Table ^C, Supporting Information) except for 2 molidustat-treated cats (Table ⁴). A slight increase in liver enzyme activity was noted for 1 cat on Day 28 and mild leukocytosis, with absolute and relative neutrophilia, absolute and relative lymphocytopenia, relative monocytopenia, and absolute eosinophilia were noted in another cat on Day 7 without further investigation or treatment (Table ⁴). Body weight measurements did not change over time, and SBP means varied between 139.4-150.8 mm Hg for the CP group and 143.9-157.3 mm Hg for the molidustat group (Figure ³).

TABLE 4

Summary of number of cats with adverse events (AEs) reported during a 28-day treatment period with either molidustat or the control product (CP), listed in descending order of frequency.

Adverse event	CP (n = 6)	Molidustat (n = 15)
Vomiting	0	6
Decreased appetite	0	2
Death	1	0
Respiratory distress	1	0
Thrombosis	0	1
Increased liver activity	0	1
Lateral recumbency	0	1
Renal lymphoma	0	1
Renal failure	0	1
Dehydration	0	1
Polyuria/polydipsia	0	1
Stomatitis	0	1
Increased CK activity ^a	0	1
Leukocytosis/neutrophilia/lymphopenia/monocytopenia/eosinophilia ^b	0	1
Lethargy	0	1
Cystitis	0	1
Mild agitation	0	1
Ear discharge	0	1

Adverse event	CP (n = 6)	Molidustat (n = 15)
Skin scab	0	1
Abnormal test result (coccidiosis)	0	1

Abbreviations: ALP, alkaline phosphatase; ALT, alanine transaminase; AST, aspartate transaminase; CK, creatine kinase; EU, Europe; RI, reference interval; WBC, white blood cells.

a

One cat (Case ID #1) with ALP 83 U/L (RI: 0-73); ALT 1058 U/L (RI: 0-175); AST 558 U/L (RI: 0-71); CK 3923 U/L (RI: 0-542) on Day 28.

b

One cat (Case ID #19) with WBC $20.6 \times 10^3/\mu\text{L}$ (RI: 3.5-16.0); absolute neutrophils $17.9 \times 10^3/\mu\text{L}$ (RI: 2.5-8.5) and relative neutrophils 87.1% (RI: 35-75); absolute lymphocytes $1.07 \times 10^3/\mu\text{L}$ (RI: 1.2-8) and relative lymphocytes 5.2% (RI: 20-45); relative monocytes 0.7% (RI: 1-4); absolute eosinophils $1.29 \times 10^3/\mu\text{L}$ (RI: 0-1) on Day 7.

FIGURE 3

Mean body weight measurements ($\text{kg} \pm \text{SD}$) and mean SBP measurements ($\text{mm Hg} \pm \text{SD}$) of anemic cats with CKD over a 28-day treatment phase with either molidustat (dotted lines) or CP (full lines) and for an additional 8-week molidustat treatment during the continuation phase. CKD, chronic kidney disease; CP, control product; SBP, systemic blood pressure.

Vomiting was the most frequently reported AE, occurring in 6 of 15 (40%) cats in the molidustat group. The number of vomiting events was 17, 9 of which came from 1 cat with decreased appetite during the first week of study. Of the other 5 cats, 3 vomited once, 1 vomited twice, and 1 vomited once 3 consecutive days during the second week of the study. None of the CP cats vomited (Table ⁴).

Three serious AEs occurred during the study. One CP cat showed respiratory distress and died on the way to the hospital (Day 14), 1 molidustat-treated cat was euthanized because of end-stage renal failure (Day 27), and another molidustat-treated cat was presented in lateral recumbency with a cold front leg and was euthanized without necropsy (Day 28; Table ⁴).

3.4

Continuation phase

Continuation phase results are presented in File ^{S1} and in Figures ^{2, 3, 4}.

FIGURE 4

Mean ($\pm \text{SD}$) PCV of 8 anemic cats with CKD treated with molidustat sodium during a 28-day treatment phase and for an additional 8 weeks during the continuation phase. CKD, chronic kidney disease.

4

DISCUSSION

We investigated the ability of molidustat, a novel HIF-PH inhibitor, to stimulate production of erythrocytes in client-owned CKD cats with anemia. Results showed the potential of molidustat to improve HCT and PCV in affected cats when compared to baseline values as early as 14-21 days after the start of daily PO administration of 5 mg/kg molidustat sodium. Although significant differences in reaching clinically relevant increases in HCT or PCV compared to CP cats were not consistently achieved, the number of individual treatment successes on Day 28 was higher among molidustat-treated cats (7/14) compared to CP cats (1/5). This lack of significant difference may be attributed to the limited sample size of the study. The number of cats achieving treatment success increased during the continuation phase. Although lack of EPO is considered to be a major factor in the anemia of CKD, other possible contributing factors potentially present in these client-owned cats may have included comorbid conditions causing inflammation, undiagnosed hyperthyroidism, coadministration of medications, or insufficient iron stores (functional or absolute). These factors may have slowed or blunted the erythropoietic response. In addition, because CKD staging was not performed pretreatment, its impact on study results is unknown. Molidustat's in-use safety profile was demonstrated in these CKD cats after up to 12 weeks of use when administered alone or in combination with other medications needed to alleviate clinical signs of CKD.

Production of EPO progressively decreases in failing kidneys because of the continued loss of functioning REP cells to fibrosis, uremic substances causing oxidative stress, chronic presence of inflammatory cytokines, and impaired iron mobilization.^{1, 8, 9, 10, 11} Overexpression of HIF inconsistently has been associated with renal fibrosis, leading to concern that HIF-PH inhibitors may promote renal fibrosis. However, studies in mice have shown that HIF-PH inhibitors do not increase renal fibrosis despite HIF stabilization and increased EPO production in pericytes.¹² Erythrocyte-stimulating agents (ESAs) such as recombinant human EPO are used as EPO replacements for anemic cats.^{1, 13, 14} This parenteral treatment has been an important tool in the management of anemia in cats. However, numerous complications such as iron deficiency, hypertension, arthralgia, fever, seizures, polycythemia, and life-threatening pure red cell aplasia have been reported in treated cats.^{1, 2, 3} Our results confirm that the EPO control mechanism is preserved in the remaining functioning cells of failing kidneys of CKD cats, and that HCT and PCV are adequate quantitative surrogate parameters to measure the erythropoietic response to daily administration of molidustat in a clinical setting. Molidustat also was well tolerated without affecting SBP in cats during our study. The use of different SBP devices at different sites and antihypertensive medications in several cats may have impacted this latter finding. A larger and longer study would provide more information regarding SBP.

Molidustat, a novel HIF-PH inhibitor, administered once daily for 28 days at 5 mg/kg significantly increased HCT and PCV as early as 14-21 days after the start of treatment compared to baseline in anemic cats with CKD.

Limitations of our study include low power (small sample size) and short duration, lack of CKD staging at baseline and unknown impact of CKD stage on study results, use of different laboratories (Europe and United States) for clinical pathology assessment, data missing for several cats at critical timepoints and use of concomitant medications in the presence of comorbidities. Despite these limitations, our objectives of investigating the ability of molidustat to safely stimulate erythropoiesis in failing kidneys of anemic cats were fulfilled. Future studies evaluating the use of molidustat in a larger number of anemic CKD cats, using HCT (or PCV) as surrogate parameters to conveniently monitor erythropoietic response, are necessary because molidustat potentially could be an innovative therapeutic option to address an unmet clinical need while safely improving the quality of life of anemic cats with CKD.

CONFLICT OF INTEREST DECLARATION

Dr S. Charles, Dr R. Süssenberger, and T. Settje are or were employed by Elanco. Drs C. Langston and C. Lainesse received payments as independent consultants.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

This study was conducted in accordance with guidelines of the Animal Care Review Committee of Bayer Animal Health (Protocol #205375) and with informed owner consent.

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

Supporting information

File S1. Continuation phase—molidustat in CKD cats.

Table A: Comorbidities and concomitant medications administered to study cats.

Table B: Summary statistics for RBC parameters of interest.

Table C: Summary statistics for biochemistry parameters.

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